Provenance Tier	Source Provenar	Confidence Scor	Assigned Matrix Weight (W_t)	
Tier 1	Primary Public R	0.95 – 1.00	0.35	
Tier 2	Archival Investiga	0.75 – 0.90	0.25	
Tier 3	Moderated Local	0.45 – 0.65	0.25	
Tier 4	Unmoderated Co	0.10 – 0.35	0.15	

Technology	Physical Property	Primary Forensic	Fundamental Physics Limit / Constraint
SAR	Microwave backs	Detect and map	Limited subsurface penetration in conductive media; clutter distortion
Dual-Comb Lase	Optical phase, fr	High-precision e	Requires direct, unimpeded optical line-of-sight to target surface
Sub-Picosecond	Photon propagati	Micro-scale surfa	Timing precision limited by pulse width and medium group refractive index (n_g)
Two-Photon Micr	Nonlinear optical	Microscopic exar	Non-destructive, but strictly limited to accessible, micro-scale surface areas
LIBS / XRF Spec	Element-specific	Identify exact ele	Requires direct optical or X-ray access to an unmasked sample surface
THz Spectroscop	Frequency-depe	Examine coating	High atmospheric attenuation; limited spatial penetration depth
GPR	Subsurface elect	Map conductive	Severe signal attenuation in high-conductivity media (e.g., wet clay); depth trade-off
LiDAR	Laser pulse ToF	Generate bare-e	Captures surface topography only; cannot directly observe buried elemental composition

Evaluation Paran	Candidate Hypot	Candidate Hypothesis: Brass			
Electrical Conduc	High ($3.5 \times	Variable by alloy ($1.2 \times 10^7 \text{ S/m} - 1.6 \times 10^7 \text{ S/m}$)			
Magnetic Respor	Non-magnetic (W	Non-magnetic (Weakly diamagnetic)			
Elemental Signat	Predominantly Al	Primary Copper (Cu) + Zinc (Zn)			
Surface Oxidatio	Thin Aluminum C	Copper/Zinc corrosion products (Verdigris/Patina)			
Optical Edge Ma	High reflectivity if	Variable optical response based on tarnishing			
SAR-Only Identif	Inconclusive (Ind	Inconclusive (Indicates conductive reflector only)			

Target Depth Res	Equivalent Roun	Equivalent Vacuum Path Length ($2 \Delta R$)	
$0.1 \text{ }\mu\te	0.667 fs	$0.2 \text{ }\mu\text{m}$	
$4.6 \text{ }\mu\te	$30.688 \text{ fs}	$9.2 \text{ }\mu\text{m}$	
$20.0 \text{ }\mu\	$133.428 \text{ fs	$40.0 \text{ }\mu\text{m}$	

Masking Parame	Urban Catchmen	Sub-Alpine / Park Catchment (e.g., STEPPACH-2015-AR)	
Conduit Architect	Concrete Utility V	Subterranean Concrete Park Drainage Pipe	
Scent Masking P	Sub-Slab Anoxic	Horizontal Air-Duct Wind-Tunnel Chimney Effect	
Vertical Plume Le	0.0% (Overric	0.0% (Carried downstream by internal airflow)	
Horizontal Ventin	Up to $192.9 \te	21.3 meters	
Primary Detectio	Borescope / $2.(	Dynamic Robotic Pipe-Crawler Scope ($60–70 \text{ ft}$ insertion)	

Case ID	Primary Subject	Target Structural	Modeled Depth	Primary Sensing	Key Taphonomic	Status / Confidence Score
STEPPACH-201	Ebby Steppach	Subterranean Co	1.2 m	Robotic Pipe-Cra	Ebby Steppach E	Recovered May 2018 (0.985)
LANDRY-2020-T	Jason Landry	Roadside Draina	0.9 m	Thermal Boresco	Culvert drainage	Active Target (0.925)
SHAFFER-2006-	Brian Shaffer	Sub-Pavement U	1.15 m	Borescope / $2.(	Sub-Slab Anoxic	Active Target SHAFFER-001 (0.930)
SHTUKA-2018-B	Ryan Shtuka	Alpine Creek Cut	0.85 m	Bare-Earth LiDAI	Snow-bridging o\	Active Target Delta (0.915)
TOMAN-2025-Q(	Liam Toman	Outflow Creek Cl	0.95 m	Dual-Freq GPR /	Tree-well Snow I	Active Target LT-HAZARD-03 (0.940)
FARBER-2015-P	Jesse Farber	Coal Quarry Ben	140 m	2.0 GHz	Traumatic Brain I	Active Target / Fragment Found (0.950)

Soil / Sediment C	Primary PCR Inh	Inhibition Risk Le	Neutralization & Extraction Protocol				
Alpine Glacial Till	Sub-alpine tannir	Extreme	Skim Milk (0.2 g/g soil) + BSA Thermal Incubation (65°C) & PowerSoil Pro Kit				
Urban Silt-Soot M	Petroleum hydro	High	Bovine Serum Albumin (BSA, 12 mg/mL) + Skim Milk Neutralization + Silica-Column PowerSoil DNA Extraction				
Cold-Water Silt /	Cold-water organ	Moderate to High	BSA Additive Amplification + Cold-Water Silt Sediment Core Filtration				
Alluvial Silt-Clay	Humic and fulvic	High	Skim Milk (0.1 g/g soil) + BSA (10 mg/mL) Competitive Neutralization & Bead-Beating				

Architectural Lay	Production Technology Component	
1. Physics & Sigr	Python 3.10, NumPy, SciPy, Custom Electromagnetic / SAR Solvers	
2. Geospatial Pro	GDAL, Rasterio, PDAL, PROJ	
3. Numerical Acc	C++ / Rust compiled extensions, GPU CUDA pipelines	
4. Evidence Data	PostgreSQL 15 / PostGIS 3.3	
5. WORM Storag	AWS S3 Object Lock / MinIO WORM Object Storage	
6. Evidence Sign	Ed25519 PKI / Protected Hardware Security Modules (HSM)	
7. API Layer	FastAPI with strict Pydantic schema validation	
8. Execution Eng	Containerized, version-pinned Docker jobs	
9. Visualization U	Leaflet.js, Mapbox GL, WebGL 3D Point Cloud Viewers	
10. Audit & Monit	OpenTelemetry, Signed Execution Manifests	
11. Deployment	Multi-stage Docker, Docker-Compose, GitHub Actions CI/CD	